Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2025-P1-Model)** **Date: 15-04-2026**

Time: 09.00Am to 12.00Pm **GTA-24** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**C**	2	**C**	3	**A**	4	**B**	5	**BD**	6	**BC**
7	**ABCD**	8	**3**	9	**3.5**	10	**4**	11	**72**	12	**3**
13	**588**	14	**D**	15	**C**	16	**A**				

PHYSICS

17	**C**	18	**A**	19	**B**	20	**A**	21	**BD**	22	**AC**
23	**ABC**	24	**9**	25	**2**	26	**625**	27	**3.40 - 3.43**	28	**8.40 - 8.58**
29	**1**	30	**A**	31	**D**	32	**D**				

CHEMISTRY

33	**C**	34	**A**	35	**D**	36	**D**	37	**BC**	38	**ACD**
39	**CD**	40	**3**	41	**4**	42	**12.5**	43	**244**	44	**19**
45	**6**	46	**D**	47	**A**	48	**B**				

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SOLUTIONS

MATHEMATICS

01. Let coordinate at point of intersection of normals at P and Q be (h,k)

Eq. of normal to the hyperbola $\frac{x^2}{3^2}-\frac{y^2}{2^2}=1$ at point $P(3\sec\theta,2\tan\theta)$ is

$3x\cos\theta+2y\cot\theta=3^2+2^2.....(1)$

Similarly, equation of normal to the hyperbola $\frac{x^2}{3^2}-\frac{y^2}{2^2}$ at point $Q(3\sec\phi,2\tan\phi)$ is

$3x\cos\phi+2y\cot\phi=3^2+2^2.....(2)$

Given $\theta+\phi=\frac{\pi}{2}\Rightarrow\phi=\frac{\pi}{2}-\theta$ and these passes through (h,k)

$\Rightarrow 3h\sin\theta+2k\tan\theta=3^2+2^2.....(3)$

And $3h\cos\theta+2k\cot\theta=3^2+2^2.....(4)$

Solving (3) and (4), we get

$K=\frac{-39(\cos\theta-\sin\theta)}{6(\cos\theta-\sin\theta)}$

Hence, ordinate of point intersection of normals at P and Q is $\frac{-13}{2}$

02. Multiply N^r & D^r by x^2

$\int\frac{x^5+x^4+x^3}{\sqrt{10x^6+12x^5+15x^4}}dx$

Put $10x^6+12x^5+15x^4=t^2$

$\int\frac{dt}{30}=\frac{t}{30}+c=\frac{x^2}{30}\sqrt{15+12x+10x^2}+c$

$g(x)=x^2$

03. Let $I=\int_0^{\pi/2}\sin\ln(\sin x)\,dx=\frac{1}{2}\int_0^{\pi/2}\sin x\ln(\sin)^2\,dx$

$\Rightarrow I=\frac{1}{2}\int_0^{\pi/2}\sin\ln(1-\cos^2 x)dx$

Put $\cos x=t\Rightarrow I=\frac{1}{2}\int_0^1\ln(1-t^2)\,dt$

$\Rightarrow I=\frac{1}{2}\left(-t^2-\frac{(t^2)^2}{2}+\frac{(-t^2)^3}{3}.....\right)dt$

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$$\Rightarrow I=-\left[\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{5}+\frac{1}{6}-\frac{1}{7}+....\right]$$

$$I=\ln\left(\frac{2}{e}\right)$$

04. The centres of the two circle will lie on the line through $P(1,2)$ perpendicular to the common tangent $4x+3y=10$. If C_1 and C_2 are the centres of these circles, then $PC_1=5=r_1$, $PC_2=-5=r_2$. Also C_1,C_2 lie on the line $\frac{x-1}{\cos\theta}=\frac{y-2}{\sin\theta}=r$,

Where $\tan\theta=\frac{3}{4}$. When $r=r_1$ the coordinates of C_1 are $(5\cos\theta+1,5\sin\theta+2)$ and $\cos\theta=\frac{4}{5}$, $\sin\theta=\frac{3}{5}$

When $r=r_2$, the coordinates of C_2 are $(-3,-1)$.

The circle with centre $C_1(5,5)$ and radius 5 touches both the coordinate axes and hence lies completely in the first quadrant

Therefore, the required circle is with centre $(-3,-1)$ and radius 5, so its equation is

$$(x+3)^2+(y+1)^2=5^2$$

$$\Rightarrow x^2+y^2+6x+2y-15=0$$

Since the origin lies inside the circle a portion of the circle lies in all the quadrants

05. Make category of points as (even, odd, odd)

Which can be done in $2\times2\times2$ i.e., ways

and in each category there are $5\times5\times5=125$ points. So, total number of ways

$=8\times{}^{125}C_2=62000=2^4\times5^3\times31$

so, the number of factors of N is $(1+4)(1+3)(1+1)=40$

06. $g'(x)=f(x)$ .....(1)

tangent to $y=f(x)$:

$y-f(x_1)=f'(x_1)(x-x_1)$ .....(2)

tangent to $y=g(x)$:

$y-g(x_1)=g'(x_1)(x-x_1)$ .....(3)

For (2) and (3), at $y=0$

$$\frac{f(x_1)}{g(x_1)}=\frac{f'(x_1)}{g'(x_1)} \quad \Rightarrow \frac{f'(x_1)}{f(x_1)}=\frac{g'(x_1)}{g(x_1)}$$

$$\Rightarrow \ln f(x)=ln(g(x))+ln\,c \quad \Rightarrow \frac{f(x)}{g(x)}=c$$

Put $x=0$, we get $c=\frac{1}{4}$ $\Rightarrow f(x)=\frac{g(x)}{4}$

$$g'(x)=\frac{g'(x)}{4}\Rightarrow g(x)=4e^{\frac{x}{4}} \quad \Rightarrow f(x)=e^{x/4}$$

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B) $I=\int_{-1}^{1}\frac{x^2+4}{4+4e^{x/4}}dx$

Apply king,

$I=\int_{-1}^{1}\frac{x^2+4}{4+4e^{-x/4}}dx$

Add $2I=\int_{-1}^{1}\frac{x^2+4}{4}dx \quad \Rightarrow I=\frac{13}{12}$

C) $h(x)=f(x)-g(x)$

$h(x)=e^{\pi/4}-4e^{x/4}$

$h'(x)=e^{x/4}\left(\frac{1}{4}-1\right) \quad (-ve)$

$\therefore h(x)$ is decreasing

07. $x^{10}+(13x-1)^{10}=0 \quad \Rightarrow x^{10}=-(13x-1)^{10}$

$\Rightarrow\left(\frac{13x-1}{x}\right)^{10}=-1\Rightarrow 13-\frac{1}{x}=(-1)^{\frac{1}{10}}$

$\Rightarrow\frac{1}{x}=13-(-1)^{\frac{1}{10}}=13-cis\left(\frac{2k\pi+\pi}{10}\right),$

$k=0,1,2,...,9$

$\frac{1}{z_1}=13-cis\left(\frac{\pi}{10}\right),\frac{1}{\bar{z}_1}=13-cis\left(\frac{-\pi}{10}\right) \Rightarrow\frac{1}{|z_1|^2}=170-26\cos\frac{\pi}{10}$

$\therefore N=\sum_{r=1}^{5}\frac{1}{|z_1|^2}=5(170)-26\left(\cos\frac{\pi}{10}+\cos\frac{3\pi}{10}+\cos\frac{5\pi}{10}+\cos\frac{7\pi}{10}+\cos\frac{9\pi}{10}\right)$

$=850\,(If\ \alpha+\beta=\pi\ then\ \cos\alpha+\cos\beta=0)$

08. Solve $y^2=2x$ and $\frac{x^2}{9}+\frac{y^2}{4}=1$ we get $P=\left(\frac{3}{2},\sqrt{3}\right),Q=\left(\frac{3}{2},-\sqrt{3}\right)$ tangent at P&Q to ellipse is $\frac{x}{6}+\frac{y}{4}\left(\sqrt{3}\right)=1$ which intersect at $R(6,0)$ tangent at $P\,\&\,Q$ to parabola $y^2=2x$ will be $y\left(\pm2\sqrt{3}\right)=\left(x+\frac{3}{2}\right)$ which intersect at $\left(-\frac{3}{2},0\right)=S$

$A=\frac{1}{2}\left(6+\frac{3}{2}\right)\times2\sqrt{3}=\frac{15\sqrt{3}}{2}$

09. Let vertices $A_1,A_2,....A_7$ are 7th roots of unity. Let $A_1(1),A_2(\alpha),A_3\left(\alpha^2\right),A_4\left(\alpha^3\right)$

$A_5=\left(\alpha^4\right)=\overline{\alpha^3},A_6=\left(\alpha^5\right)=\overline{\alpha^2},A_7=\left(\alpha^6\right)=\bar{\alpha}$

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$$(1-\alpha)\left(1-\alpha^2\right)\left(1-\alpha^3\right)\left(1-\alpha^4\right)\left(1-\alpha^5\right)\left(1-\alpha^6\right)=7$$

$$\Rightarrow\left(\left|1-\alpha\right|\left|1-\alpha^2\right|\left|1-\alpha^3\right|\right)^2=7$$

$$p_1=(A_1A_2)(A_1A_3)(A_1A_4)(A_1A_5)(A_1A_6)(A_1A_7)$$

$$=\left(\left|1-\alpha\right|\left|1-\alpha^2\right|\left|1-\alpha^3\right|\right)^2=\left(\sqrt{7}\right)^2$$

Similarly

$$p_2=\left|1-\alpha\right|\left|1-\alpha^2\right|\left|1-\alpha^3\right|\left|1-\alpha^2\right|$$

$$\sqrt{7}\left(\left|1-\alpha^3\right|\left|1-a^2\right|\right) \qquad p_3=\sqrt{7}\left(\left|1-\alpha^2\right|\right)$$

$$p_4=\sqrt{7},\ p_5=\left|1-\alpha\right|\left|1-\alpha^2\right| \text{ and } p_6=\left|1-\alpha\right|$$

$$\Rightarrow p_1\cdot p_2\cdot p_3\cdot p_4\cdot p_5\cdot p_6=\left(\sqrt{7}\right)^7$$

10. $$L=\lim_{n\to\infty} n^{\frac{-1}{2}\left(1+\frac{1}{n}\right)}\left(1^1.2^2.3^3....n^n\right)^{\frac{1}{n^2}}$$

$$\log L=\lim_{n\to\infty}\log\left[n^{-\frac{1}{2}\left(1+\frac{1}{n}\right)}\left(1^1.2^2.3^3...n^n\right)^{\frac{1}{n^2}}\right]$$

$$=\lim_{n\to\infty}\frac{-1}{2}\left(1+\frac{1}{n}\right).\log n+\log\left(1^1.2^2...n^n\right)^{\frac{1}{n^2}}$$

$$=\lim_{n\to\infty}\frac{-1}{2}\left(1+\frac{1}{n}\right).\log n+\frac{1}{n^2}\left(\log\left(1^1\right)+\log\left(2^2\right)+.....\log\left(n^n\right)\right)$$

$$=\lim_{n\to\infty}\frac{-1}{2}\left(1+\frac{1}{n}\right).\log n+\frac{1}{n^2}\left[\sum_{k=1}^{n}k\log k\right]$$

$$=\lim_{n\to\infty}\frac{-1}{2}\left(1+\frac{1}{n}\right).\log n+\frac{1}{n^2}\left[\sum_{k=1}^{n}k\log k-k\log n+k\log n\right]$$

$$=\lim_{n\to\infty}\frac{-1}{2}\left(1+\frac{1}{n}\right).\log n+\frac{1}{n^2}\left[\sum_{k=1}^{n}k\log\left(\frac{k}{n}\right)+k\log n\right]$$

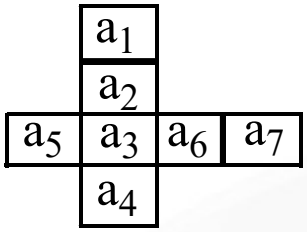
$$=\lim_{n\to\infty}\frac{-1}{2}\left(1+\frac{1}{n}\right).\log n+\lim_{n\to\infty}\frac{1}{n}.\sum_{k=1}^{n}\left(\frac{k}{n}\right)\log\left(\frac{k}{n}\right)+\lim_{n\to\infty}\frac{\log n}{n^2}\sum_{k=1}^{n}k$$

$$=\lim_{n\to\infty}\frac{-1}{2}\left(1+\frac{1}{n}\right)\log n+\int_0^1 x\ln x\,dx+\frac{\log n}{n^2}\left(\frac{n(n+1)}{2}\right) \qquad =\int_0^1 x\ln x\,dx=\frac{-1}{4}$$

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$L = e^{-1/4} = e^{-1/k} \Rightarrow k = 4$

11.



$(a_1 + a_2 + a_3 + a_4) = 21,$

$a_5 + a_3 + a_6 + a_7 = 21,$

$a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7 = 35$

$\Rightarrow a_3 = 7, a_1 + a_2 + a_4 = 14, a_5 + a_6 + a_7 = 14$

$a_1, a_2, a_4 \in \{2,4,8\}, a_5, a_6, a_7 \in \{3,5,6\}$

$a_1, a_2, a_4 \in \{3,5,6\}, a_5, a_6, a_7 \in \{2,4,8\}$

Required numbers of ways $= (3! \times 3!) + (3! \times 3!) = 72$

12. $\{1, 2, 3, \ldots\ldots, 2019\}$

$Amp\left(\frac{a-b}{a-c}\right) = 0 \Rightarrow \frac{a-b}{a-c} > 0$

We choose 3 numbers from given set and take the largest as a and other two as b and c, then number of ways

$= {}^{2019}C_3 \times 2! \times 2!$ (same for least)

$\therefore$ Required probability

$= \frac{{}^{2019}C_3 \times 2! \times 2!}{{}^{2019}C_3 \times 3!} = \frac{2}{3}$

13. Let our polynomial be $P(x)$.

It is clear that the coefficient of x in $P(x)$ is $-1+2-3+\ldots+14-15 = -8$, so $P(x) = 1 - 8x + Cx^2 + Q(x)$, where $Q(x)$ is some polynomial divisible by x^3.

Then $P(-x) = 1 + 8x + Cx^2 + Q(-x)$ and so $P(x).P(-x) = 1 + (2C - 64)x^2 + R(x)$, where $R(x)$ is some polynomial divisible by x^3.

However, we also know $P(x).P(-x) = (1-x)(1+x)(1+2x)....(1-15x)(1+15x)$

$= (1-x^2)(1-4x^2)....(1-225x^2) = 1 - (1+4+....+225)x^2 + R(x)$.

Equating coefficients, we have $2C - 64 = -(1+4+....+225) = -1240$, so $-2C = 1176$ and $|C| = \boxed{588}$.

14. $n = 50, p = \frac{1}{100}, q = \frac{99}{100}$

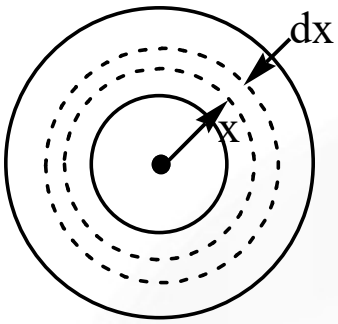
15. **CONCEPTUAL**

16. **CONCEPTUAL**

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PHYSICS

17. $dC = \dfrac{k \in_0 \times 2\pi x \ell}{dx} \quad \dfrac{2\pi \in_0 \eta \ell x^2}{dx}$



$$\frac{1}{C} = \int \frac{1}{dC} = \frac{1}{2\pi \in_0 \eta \ell} \times \int_a^b \frac{dx}{x^2}$$

$$C = 2\pi \in_0 \eta \ell \frac{(ab)}{(b-a)}$$

Capacity per unit length $= \dfrac{e}{\ell} = 2\pi\varepsilon_0\eta\left(\dfrac{ab}{b-a}\right)$

18. **CONCEPTUAL**

19. Solid angle of complete circular strip at point O,

$= 2\pi(1-\cos 53^o) - 2\pi(1-\cos 37^o) = \dfrac{2\pi}{5}5r$

$\therefore$ Solid angle between of window at point O

$\Omega = \dfrac{1}{2\pi r} \times \dfrac{2\pi}{5} = \dfrac{1}{5r}$ $\quad \therefore$ Electric flux through window

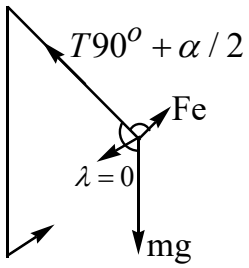
$\phi = \dfrac{\Omega}{4\pi} \times \dfrac{q}{\varepsilon_0} = \dfrac{q}{60\varepsilon_0}$ $\quad \Rightarrow \Omega = \dfrac{\pi}{15} \Rightarrow \dfrac{\ell}{5r} = \dfrac{\pi}{15} \Rightarrow \ell = \dfrac{\pi r}{3} = \dfrac{\pi}{3} \times \dfrac{6}{\pi}$ cm

$\ell = 2$ cm

20. $W = -\Delta U = \int_b^{2b} \dfrac{\mu_0 I_1 I_2}{2\pi x} \alpha dx$

21. Change in electrostatic potential energy of dipole $= \dfrac{2KP_1P_2}{r^3}$

$$\frac{Fe}{\sin\alpha} = \frac{mg}{\cos\frac{\alpha}{2}} \qquad Fe = 2mg\sin\frac{\alpha}{2}$$



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$$\frac{6kP_1P_2}{r^4}=2mg\sin\frac{\alpha}{2} \qquad KP_1P_2=\frac{mgr^4}{3}\sin\frac{\alpha}{2}$$

$$\Delta U_E=\frac{2}{3}mgr\sin\frac{\alpha}{2}=\frac{2}{3}mgh \qquad w=\frac{5}{3}mgh$$

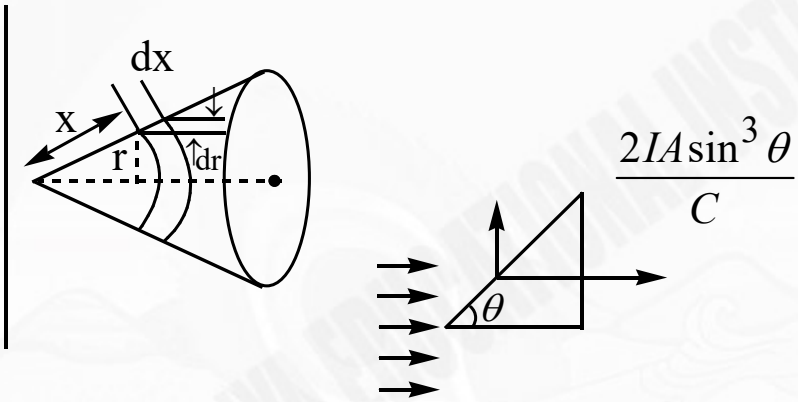
22. $\lambda_{cut\,off}=\dfrac{12400}{84000}=0.15\,\text{Å}$

$$99\%\text{ of power }=84000\times10\times10^{-3}\times0.99=300\times0.04\times4.2\frac{dT}{dt}$$

$$\Rightarrow\frac{dT}{dt}=16.5^{o}C/\sec$$

23. An ideal ammeter has zero resistance and an ideal voltmeter has infinite resistance.

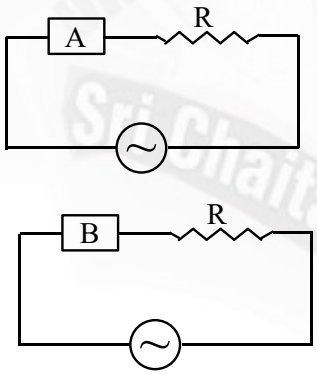
24.



$$dF_H=\frac{2I_r\cdot dA\cdot\sin^3\theta}{C} \qquad \Rightarrow\frac{2\cdot I_0r^3}{R^3}\cdot\frac{2\pi r\cdot dx\cdot\sin^2\theta\cdot\sin\theta}{C}$$

$$\Rightarrow\frac{4\pi I_0r^4dr\sin^2\theta}{R^3\cdot C} \qquad F_H=\frac{4\pi I_0\cdot R^2\cdot\sin^2\theta}{5C}$$

25.



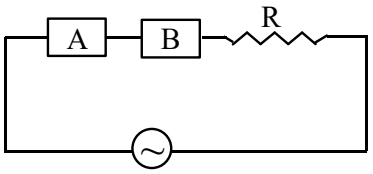
$$\phi_1=-30^0 \qquad \phi_2=+60^0$$

$$\tan\phi=\frac{X_L}{R} \qquad \tan\phi_2=\frac{X_C}{R}$$

$$X_L=\frac{R}{\sqrt{3}}$$

$$X_C=R\sqrt{3}$$

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$$\tan\theta = \frac{(X_C - X_L)}{R} = \frac{\frac{R\sqrt{3} - r}{\sqrt{3}}}{R} = \frac{2}{\sqrt{3}}$$

26. $\Delta\theta = \frac{2\lambda}{a}, \Delta\theta' = \frac{2\lambda'}{a}$

Given $\Delta\theta' = \Delta\theta + 25\%$ of $\Delta\theta$

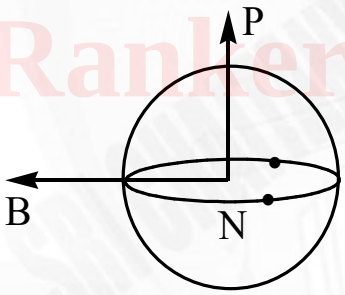
$= \Delta\theta + \frac{25}{100}\Delta\theta = 1.25\Delta\theta$

$\therefore \frac{\lambda'}{\lambda} = \frac{\Delta\theta'}{\Delta\theta} = 1.25 \Rightarrow \lambda' = 1.25\lambda = 1.25 \times 500\ nm = 625\ nm$

27. Using energy conservation law

$K = K_p + \left(m_{Li} + m_p - 2m_\alpha\right)c^2$, where K is the wanted kinetic energy K_p is the kinetic energy of the proton, m_{Li} is the mass of the lithium nucleus, m_p is the proton mass and m_α is the mass of the α par $\quad K = 3.418 MeV$

28.



Torque on the (coil + sphere) due to flow of charge through coil is $= \left|\vec{p} \times \vec{B}\right|$ (where $\vec{p}$ is the dipole moment of the coil and $\vec{B}$ is the geomagnetic field)

$= i\ N\pi\ r^2\ \ B = \int \frac{d\omega}{dt}$

$\therefore d\omega - \frac{N\pi r^2\ B}{1} idt$ or $\omega = \frac{N\pi\ r^2 B}{\frac{2}{3}mr^2}\int_0^4 idt = \frac{3\ N\pi\ B}{2}$

Ans: $\omega = \frac{3}{2}\frac{BN\pi Q}{M} = 2.7 \times 10^{-2} rad/s$

29. $Mr_1 = 2Mr_2$……..(1)

$r_1 + r_2 = r$……..(1)

$\frac{GM\,2M}{r^2} = 2MG\omega^2 . r_2$

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$\frac{GM}{r^2} = \omega^2 . \frac{r}{3}$ $\frac{3GM}{r^2} = r^2$

$\omega = \sqrt{\frac{3GM}{r^3}}$ $\lambda_2 = \frac{h}{2M . \frac{r}{3}\sqrt{\frac{3GM}{r^2}}}$

$v_2 = r_2\ \omega$ $\lambda_1 = \frac{h}{M . \frac{2r}{3}\sqrt{\frac{3GM}{r^3}}}$

$= \frac{r}{3} . \sqrt{\frac{3GM}{r^3}}$ $v_1 = \frac{2r}{3}\sqrt{\frac{3GM}{r^3}}$

ratio of $\frac{\lambda_1}{\lambda_2} = 1$

30. A) $m = +ve \Rightarrow$ v and u have the same

$\Rightarrow$It object is real image is virtual it u possible with convex lens and concave lens.

B) $m = +ve \Rightarrow$ v and u have opposite sign $\Rightarrow$ image formed u virtual

for real object concave mirror conform red (or) virtual image and for real object the convex mirror always from virtual image.

C) For real object, only convex lens can form real image

D) For mirror v & u have the same sign.

$\Rightarrow$ image formed is real. Real object real image $\Rightarrow$ concave mirror

31. In case A and C, it is similar to the combination of two capacitors of capacitances C and 4C in parallel, so electric field strength will be same $\Rightarrow \frac{E_1}{E_2} = 1$.

In case B and D, C & 4C are in series here electric field is inversely proportional to "C"

32. P) $\sqrt{D^2 + (2\lambda)^2} - D = \Delta x$

For maximas $\Delta x = n\lambda$ $4\lambda^2 = n^2\lambda_2 + 2Dn\lambda$

Only two possible value of n, n = 1, $D = \frac{3\lambda}{2}$; n = 2, D = 0

Similarly, for minima, $\Delta x = (2n-1)\frac{\lambda}{2}$

$\sqrt{D^2 + 4\lambda^2} - D = (2n-1)\frac{\lambda}{2}$

$\frac{\lambda^2}{4}\left[16 - (2n-1)^2\right] = (2n-1)\lambda D$

For D to be positive (for the right half of screen $(2n-1)^2 \le 16$

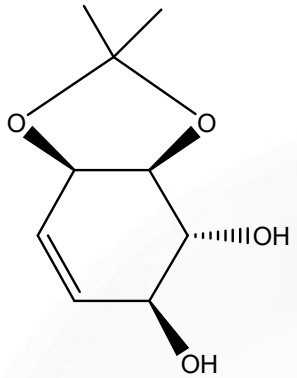
$n \le 2.5$ i.e. Two values of n possible, n = 1, 2

Similarly on the LHS at the reflecting points of n = 1, 2

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CHEMISTRY

33.

X= 

34. CONCEPTUAL

35. $V_2O_5 \xrightarrow{acids} VO_2^+$

$V_2O_5 \xrightarrow{alkalies} VO_4^{3-}$

$Cr < Mn : IE_1$

$Cr > Mn : IE_2$

$Cr < Mn : IE_3$

36. $Au \xrightarrow{Aqua\ regia} [AuCl_4]^-$

37. Phenol benzoic acid can not be differentiated by aq. NaOH.
Compound (R) upon reaction with $AcCl$ / $Pyridine$ gives acetanilide as the major product

38. Theory based

39. $(i)\left[Cu(NH_3)_4\right]^{2+} = Square\ planar - para$

$(ii)\left[Ni(CO)_4\right] = Tetrahedral - dia$

$(iii)Co^{+2} : d^7 : CN^- : SFL : t_{2g}^6 eg^1$

$(iv)\left[Co(ox)_3\right]^{3-} = Co^{3+} : d^6 : t_{2g}^6 : CFSE = -2.4\Delta_0$

$\left[Cr(ox)_3\right]^{3-} = Cr^{3+} : d^3 : t_{2g}^3 : CFSE = -1.2\Delta_0$

40. a,c,e,g,h

41. (ii, iii, iv, vi)

42.

$A \longrightarrow B + C$

$(1-x) \quad \frac{x}{3} \quad \frac{2x}{3}$

$(1-x)\times 15 + \frac{x}{3}\times 10 + \frac{2x}{3}(-20) = 0$

$x = 3/5$

$K_{overall} = 3K = 0.072$

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$$K_{overall} = \frac{1}{t}\ln\left(\frac{1}{1-x}\right)$$

On solving we get

$t = 12.5\,\text{sec}$

43. $A + B \underset{k_2}{\overset{k_1}{\rightleftharpoons}} C + D$

$$\frac{k_1}{k_2} = \frac{(C)(D)}{(A)(B)}$$

$$(C) = \frac{k_1}{k_2}\frac{(A)(B)}{(D)}$$

$2C \xrightarrow{k_2} G + H \ [R.D.S]$

$\text{Rate} = k_2[C]^2$

$$\text{Rate} = k_2\left[\frac{k_1}{k_2}\frac{(A)(B)}{(D)}\right]^2$$

$$\text{So } k = \frac{k_2.k_1}{(k_1)^2}$$

$E_a = E_{a_2} + 2E_{a_1} - 2E_{a-1}$

$= 196 + 2\times 120 - 2\times 96$

$= 196 + 48$

$= 244\,kJ$

44. $a = 8, b = 11$

$MnO_4^{2-} \longrightarrow MnO_4^- + MnO_2$

45. $X = [Ag(CN)_2]^-$, $P = [Au(CN)_2]^-$

Both X and P are diamagnetic

$Y = Q = \underset{\text{(Tetrahedral)}}{[Zn(CN)_4]^{2-}} : Zn^{2+} : d^{10} : e^4t_2^6$

46. Cellulose & Amylose gives only glucose on hydrolysis

47. Ester & acid halide each will consume 2-equalance of RMgX, aldehyde & ketones will consume 1-equalant of RMgX

48. **A)** $n_A = 1 \quad n_B = 2$

$$X_A = \frac{1}{3} \quad X_B = \frac{2}{3}$$

$P = X_A P_A^{\,0} + X_B P_B^{\,0}$

$$= \frac{1}{3}\times 150 + \frac{2}{3}\times 75$$

$= 100mm$

$P_{observed} > P_{ideal}$

So positive deviation

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B) $y_A = 0.6 \quad y_B = 0.4$

$$\frac{y_A}{y_B} = \frac{X_A.P_A^0}{X_B.P_B^0} = \frac{X_A P_A^0}{(1-X_A)P_B^0}$$

$X_A = 0.5 \quad X_B = 0.5$

$P = X_A P_A^0 + X_B P_B^0$

$P = 100mm$

$P_{observed} = P_{ideal}$

$\Rightarrow$ Solution obeys Raoult's law

C) $P = \frac{3}{5} \times 150 + \frac{2}{5} \times 150$

$= 150\,mm$

$P_{observed} < P_{ideal}$

So negative deviation

D) $P = \frac{1}{3} \times 90 + \frac{2}{3} \times 120$

$P = 110\,mm$

$P_{observed} = P_{ideal}$

$\Rightarrow$ Solution obeys Raoult's law

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